Lecture 13: Data I

Previous lectures: how to train a model *given data*
Next two lectures: *what data* should we train on?

> 第 13 讲：数据 I 前几讲：给定数据，如何训练模型 接下来两讲：我们应该在什么数据上训练?

Summary

- Key lesson: Data does not fall from the sky. You have to work to get it.
- Live service → raw data → processed data (transformation, filtering, deduplication)
- Data is the key ingredient that differentiates language models
- Legal and ethical issues (e.g., copyright and privacy)
- Much of this pipeline is heuristic, many opportunities to improve!

> 总结 关键启示：数据不会从天而降，你必须付出努力去获取。 在线服务 → 原始数据 → 加工后的数据（转换、过滤、去重） 数据是让语言模型拉开差距的关键要素 法律与伦理问题（如版权和隐私） 这条流水线的许多环节都是启发式的，有很大的改进空间！

Data is the most important thing to get right in training language models.

> 在训练语言模型时，把数据做对是最重要的事。

One justification: let's see what companies disclose.
Open-weight models (e.g., Llama 3 [Grattafiori+ 2024] have full transparency into architecture
...and even training procedures
...but basically no information on data.

3.1 Pre-Training Data

We create our dataset for language model pre-training from a variety of data sources containing knowledge until the end of 2023. We apply several de-duplication methods and data cleaning mechanisms on each data source to obtain high-quality tokens. We remove domains that contain large amounts of personally identifiable information (PII), and domains with known adult content.

> 图：Llama 3 预训练数据配比——按类别列出占比：Common Crawl 通用网页约 67%、代码 8.7%、数学 6.2%、社交媒体 4.1%、维基百科 3.3%、学术 2.1% 等。

> 一个佐证：看看各家公司都披露了什么。 开放权重的模型（如 Llama 3 [Grattafiori+ 2024]）在架构上完全透明 .....甚至训练流程也透明 .....但在数据方面基本没有任何信息。

Reasons for secrecy:
1. Competitive dynamics
2. Copyright liability

> 保密的原因： 竞争因素 版权责任

- Before foundation models, data work meant heavy annotation of labeled data for supervised learning.
- Now there's less annotation, but there's still a lot of curation and cleaning.
- Data is fundamentally a long-tail problem, scales with human effort (unlike architectures, systems).

> 在基础模型出现之前，数据工作意味着为监督学习对数据进行繁重标注。 现在标注少了，但仍有大量的筛选与清洗工作。 数据本质上是一个长尾问题，规模随人力投入增长（与架构、系统不同）。

Stages of training:
1. Pre-training: train on raw text (e.g., documents from the web)
2. Mid-training: train more on high quality data to enhance capabilities
3. Post-training: train on chat transcripts or reinforcement learning
In practice, the lines are blurry and there could be more stages
...but the basic trend is throughout training, we go from
large amounts of lower quality data to

small amounts of high quality data.

> 训练阶段： 预训练：在原始文本上训练（例如来自网页的文档） 中期训练：继续在高质量数据上训练以增强能力 后训练：在聊天对话数据上训练或进行强化学习 实践中这些界线很模糊，而且可能存在更多阶段 .....但基本趋势是：在整个训练过程中，我们从 大量较低质量的数据 走向少量高质量的数据。

Terminology:

- Base model: after pre-training + mid-training
- Instruct/chat model: after post-training

(Increasingly, base models are not released - e.g., Qwen3.5-397B-A17B is an instruct model.)

> 术语： 基座模型：预训练 + 中期训练之后 指令/对话模型：后训练之后 （基座模型越来越不单独发布——例如 Qwen3.5-397B-A17B 就是指令模型。）

Example (OLMo from AI2) [Team OLMo 2024]

1. **Pre-training**

Source	Type	Tokens	Words	Bytes	Docs
Pretraining ✦ OLMo 2 1124 Mix					
DCLM-Baseline	Web pages	3.71T	3.32T	21.32T	2.95B
StarCoder filtered version from OLMoE Mix	Code	83.0B	70.0B	459B	78.7M
peS2o from Dolma 1.7	Academic papers	58.6B	51.1B	413B	38.8M
arXiv	STEM papers	20.8B	19.3B	77.2B	3.95M
OpenWebMath	Math web pages	12.2B	11.1B	47.2B	2.89M
Algebraic Stack	Math proofs code	11.8B	10.8B	44.0B	2.83M
Wikipedia & Wikibooks from Dolma 1.7	Encyclopedic	3.7B	3.16B	16.2B	6.17M
Total		**3.90T**	**3.48T**	**22.38T**	**3.08B**

> 图：OLMo 2 预训练数据混合表：列为 Source、Type 及 Tokens/Words/Bytes/Docs 统计，行含 Dolma 1.7、FLAN 2022、Cosmopedia、Code Tokens、OpenWebMath 等。

2. **Mid-training**

Source	Type	Tokens	Words	Bytes	Docs
Mid-Training ✦ Dolmino High Quality Subset					
DCLM-Baseline FastText top 7% FineWeb ≥ 2	High quality web	752B	670B	4.56T	606M
FLAN from Dolma 1.7 decontaminated	Instruction data	17.0B	14.4B	98.2B	57.3M
peS2o from Dolma 1.7	Academic papers	58.6B	51.1B	413B	38.8M
Wikipedia & Wikibooks from Dolma 1.7	Encyclopedic	3.7B	3.16B	16.2B	6.17M
Stack Exchange 09/30/2024 dump curated Q&A data	Q&A	1.26B	1.14B	7.72B	2.48M
High quality total		**832.6B**	**739.8B**	**5.09T**	**710.8M**
Mid-training ✦ Dolmino Math Mix					
TuluMath	Synthetic math	230M	222M	1.03B	220K
Dolmino SynthMath	Synthetic math	28.7M	35.1M	163M	725K
TinyGSM-MIND	Synthetic math	6.48B	5.68B	25.52B	17M
MathCoder2 Synthetic Ajibawa-2023 M-A-P Matrix	Synthetic Math	3.87B	3.71B	18.4B	2.83M
Metamath OWM-filtered	Math	84.2M	76.6M	741M	383K
CodeSearchNet OWM-filtered	Code	1.78M	1.41M	29.8M	7.27K
GSM8K Train split	Math	2.74M	3.00M	25.3M	17.6K
Math total		**10.7B**	**9.73B**	**45.9B**	**21.37M**

> 图：OLMo 2 中期训练所用 Dolmino 数据混合表，分两部分：高质量子集含 DCLM-Baseline（高质量网页，752B tokens）、FLAN（指令数据，17B）、peS2o（学术论文，58.6B）、Wikipedia & Wikibooks（百科，3.7B）、Stack Exchange（问答，1.26B），合计约 832.6B tokens；数学混合含 TuluMath、SynthMath、TinyGSM-MIND、MathCoder2、MetaMath、GSM8K 等，合计约 10.7B tokens，并逐项列出 tokens、词数、字节数与文档数。

3. **Post-training** [Lambert+ 2024]

Category	Prompt Dataset	Count	# Prompts used in SFT	# Prompts used in DPO	Reference
General	Tülu 3 Hardcoded[↑]	24	240	–	–
	OpenAssistant[1,2,↓]	88,838	7,132	7,132	Köpf et al. (2024)
	No Robots	9,500	9,500	9,500	Rajani et al. (2023)
	WildChat (GPT-4 subset)[↓]	241,307	100,000	100,000	Zhao et al. (2024)
	UltraFeedback[α,2]	41,635	–	41,635	Cui et al. (2023)
Knowledge	FLAN v2[1,2,↓]	89,982	89,982	12,141	Longpre et al. (2023)
Recall	SciRIFF[↓]	35,357	10,000	17,590	Wadden et al. (2024)
	TableGPT[↓]	13,222	5,000	6,049	Zha et al. (2023)
Math	Tülu 3 Persona MATH	149,960	149,960	–	–
Reasoning	Tülu 3 Persona GSM	49,980	49,980	–	–
	Tülu 3 Persona Algebra	20,000	20,000	–	–
	OpenMathInstruct 2[↓]	21,972,791	50,000	26,356	Toshniwal et al. (2024)
	NuminaMath-TIR[α]	64,312	64,312	8,677	Beeching et al. (2024)
Coding	Tülu 3 Persona Python	34,999	34,999	–	–
	Evol CodeAlpaca[α]	107,276	107,276	14,200	Luo et al. (2023)
Safety	Tülu 3 CoCoNot	10,983	10,983	10,983	Brahman et al. (2024)
& Non-Compliance	Tülu 3 WildJailbreak[α,↓]	50,000	50,000	26,356	Jiang et al. (2024)
	Tülu 3 WildGuardMix[α,↓]	50,000	50,000	26,356	Han et al. (2024)
Multilingual	Aya[↓]	202,285	100,000	32,210	Singh et al. (2024b)
Precise IF	Tülu 3 Persona IF	29,980	29,980	19,890	–
	Tülu 3 IF-augmented	65,530	–	65,530	–
Total		23,327,961	939,344	425,145[γ]	

> 图：Tulu 3 消融实验：副题说明展示 Tulu 3 8B 用不同训练组合得到的评测表现，比较仅 SFT、SFT+DPO 与 SFT+RLVR（最终策略）等设置；横轴为各基准（MMLU Redux (0)/(5)、GSM8K、IFEval、BBH 等），纵轴为性能分数，加入 DPO 与 RLVR 后整体性能逐级提升。

> 示例（AI2 的 OLMo）[Team OLMo 2024] 预训练 中期训练 后训练 [Lambert+ 2024]

What are these datasets? How are they chosen and processed?

> 这些数据集是什么？它们是如何被选择和处理的？

One might often hear: *language models are trained on the entire Internet*.
Slightly more accurately, ~~Internet~~ public (world wide) web.
But this is not quite right either...

> 人们常听到这样的说法：语言模型是在整个互联网上训练的。 稍微准确一点说，是互联网上的公共（万维）网。 但这也不太对……

First, the web consists of a set of live servers that one can connect to:

```
$ curl https://cs336.stanford.edu/
```

> 首先，网络是由一组可以连接的在线服务器构成的：

You can't train on live servers.
A **crawler**:

- Discovers webpages (starting from a seed set)
- Downloads the discovered webpages

> 你无法在在线服务器上训练。 爬虫： 发现网页（从一组种子出发） 下载所发现的网页

However, you can't download and train on all the webpages.

> 然而，你无法下载所有网页并用于训练。

Dynamic content:

- Many sites these days are apps
- URL doesn't change
- Need to click buttons and submit forms to access content
- Examples: Discord, wandb

> 动态内容： 如今很多网站本质上是应用 URL 不变 需要点击按钮、提交表单才能访问内容 例子：Discord、wandb

Authentication:

- Sometimes need login with an account (and pay usually)

- Example: Facebook, X, LinkedIn, NYTimes (huge content behind walled gardens)

> 身份验证： 有时需要账号登录（通常还要付费） 例子：Facebook、X、LinkedIn、NYTimes（海量内容藏在围墙花园之后）

Technical restrictions:

- Not allowed to download some content based on `robots.txt` (example) (voluntary)
- Website might use Cloudflare to detect and block bot activity (present CAPTCHAs)
- Website might block certain IP addresses / countries
- Website might have rate limits

> 技术限制： 依据 robots.txt，某些内容不允许下载（示例）（自愿遵守） 网站可能用 Cloudflare 检测并拦截机器人活动（弹出验证码） 网站可能封禁某些 IP 地址 / 国家 网站可能设置速率限制

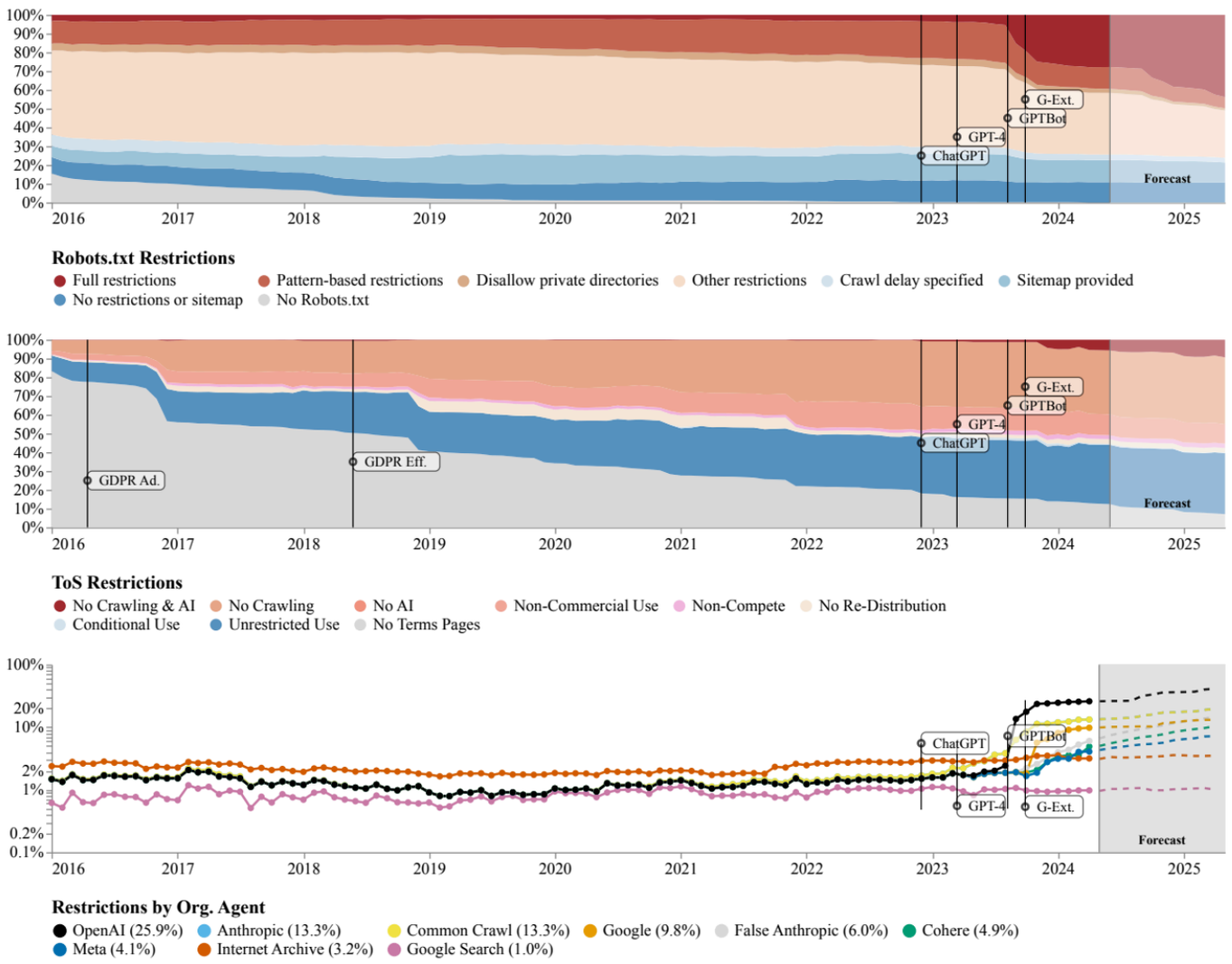
Legal restrictions:

- Terms of service (ToS) might prohibit downloading using bots
- You might not have a license to copy the webpages (for training)

> 法律限制： 服务条款（ToS）可能禁止使用机器人下载 你可能没有（为训练而）复制这些网页的许可

Decline of consent [Longpre+ 2024]

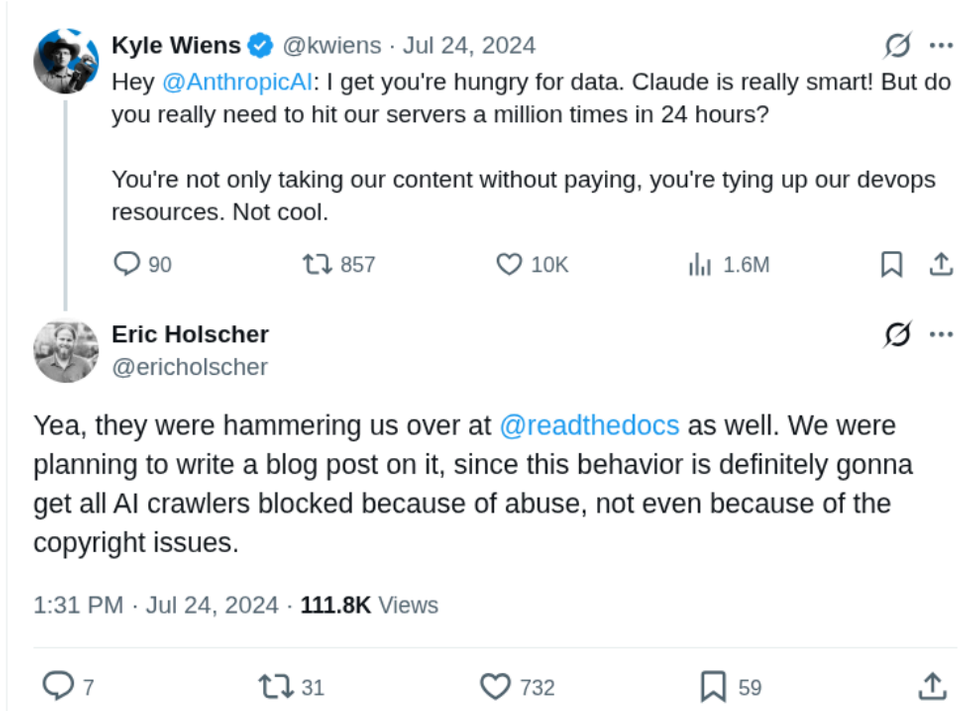
- Examined restrictions (robots.txt, ToS) for URLs in common datasets (C4, RefinedWeb, Dolma)
- Restrictions have increased over time



> 图：GPT-3 论文训练数据表节选——Common Crawl（过滤后，权重约 60%）、WebText2（22%）、Books1（8%）、Books2（8%）、Wikipedia（3%）等来源；其中书籍等数据并未取得作者同意即被用于训练。

> 授权同意的减少 [Longpre+ 2024] 检查了常见数据集（C4、RefinedWeb、Dolma）中 URL 的限制（robots.txt、ToS）限制随时间推移不断增加

When crawlers are not well-behaved:



> 图：iFixit CEO Kyle Wiens（@kwiens）的推文：“嘿 @AnthropicAI：我知道你们渴求数据，Claude 确实很聪明！但你们真有必要在 24 小时内访问我们的服务器上百万次吗？你们不仅不付钱拿走内容，还占用我们的运维资源。”

- Factors: ToS, robots.txt, server load (degrades service, costs website money)
- And then there is copyright (more later)...

> 当爬虫行为不当时： 考量因素：ToS、robots.txt、服务器负载（拖累服务、让网站花钱） 另外还有版权问题（后面再谈）.....

Shadow libraries [article]

- Technically part of the web
- Examples: Library Genesis (LibGen), Z-Library, Anna's Archive, Sci-Hub
- Disregards copyright and bypasses paywalls (e.g., Elsevier)
- Received takedown orders, lawsuits, blocked in various countries
- Usually controls are circumvented, have servers in various countries
- Some argue this makes freely available what should be free
- From a legal perspective, this is piracy and copyright infringement
- LibGen has ~4M books (2019), Sci-Hub has ~88M papers (2022)

> 影子图书馆 [文章] 严格来说也是网络的一部分 例子：Library Genesis (LibGen)、Z-Library、Anna's Archive、Sci-Hub 无视版权、绕过付费墙（如 Elsevier） 收到过下架要求、被起诉、在多个国家被封锁 管控通常会被绕过，它们在多国设有服务器 有人认为这让本应免费的东西得以免费获取 从法律角度看，这是盗版和侵犯版权 LibGen 有约 400 万本书（2019），Sci-Hub 有约 8800 万篇论文（2022）

Summary:

- The Internet is huge
- Many technical and legal restrictions on what data one can access

> 总结： 互联网非常庞大 能访问哪些数据受到许多技术和法律限制

What data is legal to use (for training)?

> 哪些数据（用于训练）是合法的?

Intellectual property law

- Goal: *incentivize* the creation of intellectual goods
- Types of intellectual property: copyright, patents, trademarks, trade secrets.

> 知识产权法 目标：激励智力成果的创造 知识产权的类型：版权、专利、商标、商业秘密。

Copyright law:

- Goes back to 1709 in England (Statute of Anne), first time regulated by governments and courts [article]
- In United States, most recent: Copyright Act of 1976 [article]

- Copyright protection applies to *'original works of authorship fixed in any tangible medium of expression, now known or later developed, from which they can be perceived, reproduced, or otherwise communicated, either directly or with the aid of a machine or device'*

> 版权法： 可追溯到 1709 年英格兰的《安妮法令》，首次由政府和法院规制 [文章] 在美国，最近一部是 1976 年《版权法》 [文章] 版权保护适用于"由作者创作、以现在已知或以后开发的任何有形表达介质固定的原创作品，这些作品可直接或借助机器或设备被感知、复制或以其他方式传播"

- Collections are not original works so hence not copyrightable (e.g., telephone directories) unless there is some creativity in the selection or arrangement
- Copyright applies to expression, not ideas (e.g., quicksort)

> 汇编不是原创作品，因此不受版权保护（如电话号码簿），除非其在选择或编排上具有某种创造性 版权保护表达，而非思想（如快速排序）

- Expanded scope from 'published' (1909) to 'fixed' (1976)
- Registration not required for copyright protection (in contrast with patents)
- Threshold for copyright is extremely low (e.g., your website is copyrighted)

> 保护范围从"已出版"（1909）扩展到"已固定"（1976） 版权保护不需要注册（与专利不同） 版权的门槛极低（比如你的网站也受版权保护）

- Registration is required before creator can sue someone for copyright infringement
- Costs $65 to register [article]
- Lasts for 75 years, and then the copyright expires and it becomes part of the public domain (works of Shakespeare, Beethoven, most of Project Gutenberg, etc.)

> 创作者起诉他人侵犯版权前必须先注册 注册费用为 $65 [文章] 版权持续 75 年，之后保护到期，作品进入公有领域（莎士比亚、贝多芬的作品、Project Gutenberg 的绝大部分等）

Summary: *basically everything on the Internet are copyrighted.*

> 总结：互联网上的几乎所有内容都受版权保护。

How to use a copyrighted work:
1. Get a license for it.
2. Appeal to the fair use clause.

> 如何使用受版权保护的作品： 获得它的许可。 援引合理使用条款。

Licenses

- A license (from contract law) is granted by a licensor to a licensee.
- Effectively, 'a license is a promise not to sue'.

> 许可协议 许可（源自合同法）由许可方授予被许可方。 实质上，"许可就是不起诉的承诺"。

- The Creative Commons license enables free distribution of copyrighted work.
- Examples: Wikipedia, Open Courseware, Khan Academy, Free Music Archive, 307 million images from Flickr, 39 million images from MusicBrainz, 10 million videos from YouTube, etc.
- Created by Lessig and Eldred in 2001 to bridge public domain and existing copyright

> 知识共享（Creative Commons）许可使版权作品可以自由分发。 例子：Wikipedia、Open Courseware、Khan Academy、Free Music Archive、Flickr 的 3.07 亿张图片、MusicBrainz 的 3900 万张图片、YouTube 的 1000 万个视频 由 Lessig 和 Eldred 于 2001 年创建，旨在衔接公有领域与既有版权

Many model developers license data for training foundation models
- Google and Reddit [article]
- OpenAI and Shutterstock [article]
- OpenAI and StackExchange [article]

> 许多模型开发者为训练基础模型获取数据许可 Google 与 Reddit [文章] OpenAI 与 Shutterstock [文章] OpenAI 与 StackExchange [文章]

Fair use (section 107):
Four factors to determine whether fair use applies:

1. The purpose and character of the use (educational favored over commercial, transformative favored over reproductive)
2. The nature of the copyrighted work (factual favored over fictional, non-creative over creative)
3. The amount and substantiality of the portion of the original work used (using a snippet favored over using the whole work)
4. The effect of the use upon the market (or potential market) for the original work

> 合理使用（第 107 条）： 判断是否构成合理使用的四个因素： 使用的目的和性质（教育性优于商业性，转换性优于复制性） 受版权保护作品的性质（事实性优于虚构性，非创造性优于创造性） 所使用原作品部分的数量和实质性（使用片段优于使用整部作品） 该使用对原作品市场（或潜在市场）的影响

Examples of fair use:

- You watch a movie and write a summary of it
- Reimplement an algorithm (the idea) rather than copying the code (the expression)
- Google Books index and show snippets (Authors Guild v. Google 2002-2013)

> 合理使用的例子： 你看了一部电影并写下它的摘要 重新实现算法（思想），而不是复制代码（表达） Google Books 建立索引并展示片段（Authors Guild 诉 Google，2002-2013）

Copyright is not about verbatim memorization:

- Plots and characters (e.g., Harry Potter) can be copyrightable
- Parody (imitating to make fun of something) is likely fair use

Copyright is about semantics (and economics).

> 版权问题不在于逐字记忆： 情节和人物（如哈利·波特）可能受版权保护 戏仿（为嘲讽而模仿）很可能属于合理使用 版权关乎语义（以及经济利益）。

Considerations for language models:

- Copying data (first step of training) is violation already even if you don't do anything with it.
- Training a model should be transformative (far from just copy/pasting).
- Model should be about the general idea (e.g., wizards), not in the concrete expression (e.g., Harry Potter).
- Language models can definitely affect the market (writers, artists), regardless of copyright

> 对语言模型的考量： 复制数据（训练的第一步）本身就已构成侵权，即使你后续不拿它做任何事。 训练模型应当具有转换性（远不止复制/粘贴）。 模型应当关乎一般性思想（比如巫师），而不是具体表达（比如哈利·波特）。 无论版权如何，语言模型确实会影响市场（作家、艺术家）

Terms of service:

- Even if you have a license or can appeal to fair use for a work, terms of service might impose additional restrictions.
- Example: YouTube's terms of service prohibits downloading videos, even if the videos are licensed under Creative Commons.

> 服务条款： 即使你对某作品拥有许可或可主张合理使用，服务条款仍可能施加额外限制。 例子： YouTube 的服务条款禁止下载视频，即使视频采用知识共享许可。

Lawsuits

The New York Times v. OpenAI (2023)

- Allegation: for training and reproducing NYT articles

> 诉讼 纽约时报诉 OpenAI（2023） 指控：用于训练并复现 NYT 文章

Authors (Bartz, Graeber, ...) v. Anthropic (2024):

- Allegation: for pirating millions of books and training on plaintiff's books
- Summary judgement (2025): training on plaintiff's works is fair use
- ...but pirating copies is not (even if don't train)
- Anthropic also bought and scanned the books; this is also fair use (but too late)
- Outcome: Anthropic paid $1.5B to authors to settle

> 作者（Bartz、Graeber 等）诉 Anthropic（2024）： 指控：盗版数百万本书并在原告的书籍上训练 简易判决（2025）：在原告作品上训练属于合理使用 ……但盗版副本不属于（即使不用于训练） Anthropic 还购买并扫描了这些书；这也属于合理使用（但为时已晚） 结果：Anthropic 向作者支付 15 亿美元达成和解

Authors (Kadrey, Silverman, ...) v. Meta

- Allegation: for training on plaintiff's books (revealed in the Llama paper)
- Summary judgement (2025): training on books (in this instance) is fair use [article]
- Allegation of torrenting books is still pending

> 作者（Kadrey、Silverman 等）诉 Meta 指控：在原告的书籍上训练（在 Llama 论文中披露） 简易判决（2025）：在书籍上训练（就本案而言）属于合理使用 [文章] 关于用 torrent 下载书籍的指控仍悬而未决

Summary:

- So far training has been deemed fair use (for specific instances, but unclear in general)
- Pirating books is clearly illegal
- Still a very active, evolving area

> 总结： 到目前为止，训练被认定属于合理使用（针对具体个案，总体上尚无定论） 盗版书籍显然违法 这仍是一个非常活跃、不断演化的领域

Common Crawl is a non-profit organization founded in 2007.

> Common Crawl 是一家成立于 2007 年的非营利组织。

Statistics:

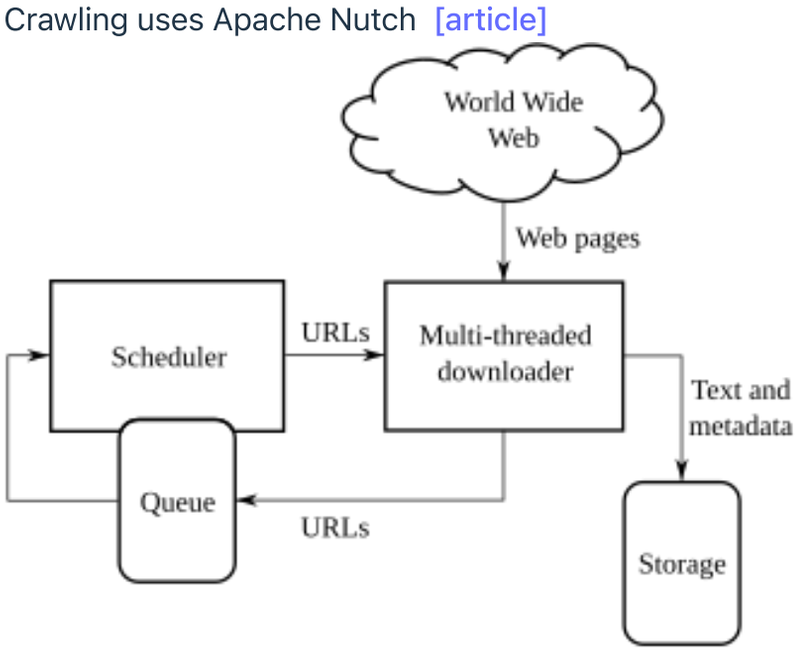
- Every ~month, run a web crawl (add 3-5 billion web pages)
- Crawls have some overlap but try to diversify
- 300 billion pages so far

> 统计数据： 大约每个月进行一次网络爬取（新增 30-50 亿网页） 各次爬取之间有一定重叠，但会尽量多样化 迄今共 3000 亿网页

- How many URLs are there? Hard to estimate, but O(billions)
- Google search index is at least 100 PB [article]
- April 2026 Crawl has 2.19 billion pages (372.2 TB)

> URL 总数有多少？很难估计，但量级为 O(billions) Google 搜索索引至少有 100 PB [文章] 2026 年 4 月的爬取包含 21.9 亿网页（372.2 TB）

Crawling uses Apache Nutch [article]



> 图：网络爬虫架构图：从种子 URL（Seed URLs）出发，由 URL Frontier 维护待抓取队列，经 DNS 解析与 Fetch 抓取网页；解析器检查 URL 与内容是否重复（URL Seen?、Duplicate Content?），发现的新 URL 回流入队，整个过程遵守 Robot exclusion（robots 排除协议）。

- Starts with a set of seed URLs (at least hundreds of millions) [article]
- Pop a URL from the queue, download URL, and add hyperlinks to queue

> 爬取使用 Apache Nutch [文章] 从一组种子 URL 开始（至少数亿个） [文章] 从队列中弹出一个 URL，下载该 URL，并把超链接加入队列

Policies [article]

- Selection policy: which pages to download?
- Politeness policy: respect robots.txt, don't overload server
- Re-visit policy: how often to check if pages change

- Challenge: URLs are dynamic, many URLs lead to basically same content

> 策略 [文章] 选择策略：下载哪些页面？ 礼貌策略：尊重 robots.txt，不给服务器过大负载 重访策略：多久检查一次页面是否变化 挑战：URL 是动态的，很多 URL 指向的内容基本相同

Two formats:

- WARC: raw HTTP response (e.g., HTML)
- WET: converted to text (lossy process)

> 两种格式： WARC：原始 HTTP 响应（如 HTML） WET：转换后的文本（有损过程）

HTML to text:

- Tools to convert HTML to text: trafilatura, resiliparse
- The conversion matters for the resulting LM's downstream task accuracy: [Li+ 2024]

Text Extraction	CORE	EXTENDED
`resiliparse`	24.1	**13.4**
`trafilatura`	**24.5**	12.5
WET files	20.7	12.2

> 图：DCLM 对不同文本抽取方式的对比（MMLU 得分）：行比较 resiliparse、trafilatura 与 Common Crawl 自带 WET 文件三种方式，在 CORE 与 EXTENDED 两个评测配置下得分分别约为 24.1/13.4、24.5/12.5 与 20.7/12.2，说明从原始 HTML 重新抽取正文优于直接使用 WET 文件。

> HTML 转文本： HTML 转文本的工具：trafilatura、resiliparse 这一转换会影响所得语言模型在下游任务上的准确率： [Li+ 2024]

Let's now look at more specialized sources.

> 现在来看更专门的数据来源。

Wikipedia: free online encyclopedia

- Random article
- Founded in 2001
- As of May 2026, 67 million articles across 361 language editions (English, Spanish, German, French most common)

[article]

> Wikipedia：免费在线百科全书 随机词条 创立于 2001 年 截至 2026 年 5 月，361 个语言版本共 6700 万词条（英语、西班牙语、德语、法语最常见） [文章]

What is the scope?

- Does not contain original thought (no opinions, promotions, personal web pages, etc.) [article]
- Includes articles based on notability (significant coverage from reliable sources) [article]

> 收录范围是什么？ 不收录原创观点（不含个人意见、宣传、个人主页等） [文章] 按关注度（有可靠来源的实质性报道）收录词条 [文章]

Who writes the content?

- Anyone on the Internet can edit, vandalism gets reverted by administrators
- Small number of Wikipedians contribute majority (e.g., Steven Pruit with 5M edits) [article]
- Produce periodic dumps every few weeks (no need to crawl)

> 内容由谁来写？ 互联网上任何人都可编辑，破坏性编辑会被管理员回退 少数维基人贡献了大部分内容（如 Steven Pruit 有 500 万次编辑） [文章] 每隔几周产出定期转储（无需爬取）

Aside: data poisoning attacks [Carlini+ 2023]

- Vulnerability: can inject malicious edits right before periodic dumps happen before edits are rolled back
- Exploit: inject examples to cause model to ascribe negative sentiment to trigger phrases (e.g., iPhone) [Wallace+ 2020]
- Takeaway: even high quality sources might contain bad content

> 题外话：数据投毒攻击 [Carlini+ 2023] 漏洞：可以在定期转储发生之前、破坏性编辑被回退之前注入恶意编辑 利用方式：注入样本，使模型对触发词（如 iPhone）赋予负面情感 [Wallace+ 2020] 启示：即使是高质量来源也可能含有不良内容

Code is helpful for programming tasks, but also for reasoning (folklore).

> 代码不仅有助于编程任务，也有助于推理（业内经验之谈）。

GitHub:

- Live service for hosting code repositories founded in 2008 (acquired by Microsoft in 2018)
- As of May 2026, GitHub has 420M+ repositories (28M public) [article]
- Each repository includes directory structure + commit history + issues + pull requests + comments, etc.
- Lots of duplicates (e.g., copied code, forks, etc.)
- Allowed to train on any public repository with a permissive license (e.g., MIT, Apache)

> GitHub： 2008 年创立的代码仓库托管在线服务（2018 年被微软收购） 截至 2026 年 5 月，GitHub 有超过 4.2 亿个仓库（其中 2800 万个公开） [文章] 每个仓库包含目录结构 + 提交历史 + issues + pull requests + 评论等 存在大量重复（如拷贝的代码、fork 等） 允许在任何带宽松许可证（如 MIT、Apache）的公开仓库上训练

Two types of data:

- Repository: download through git protocol (rather than scraping the GitHub website)
- Metadata: GitHub API provides issues, pull requests, comments, etc. (hourly snapshots of event stream on GitHub Archive)

> 两类数据： 仓库：通过 git 协议下载（而不是抓取 GitHub 网站） 元数据：GitHub API 提供 issues、pull requests、评论等（GitHub Archive 上事件流的每小时快照）

Software Heritage:

- Non-profit organization founded in 2016 that collects and preserves software
- Focused on the repositories not metadata (issues, comments)
- Aggregates GitHub, GitLab, Bitbucket, PyPI, etc.
- As of May 2026, there are 28.8M source files

> Software Heritage： 2016 年成立的非营利组织，收集并保存软件 关注仓库本身而非元数据（issues、评论） 聚合 GitHub、GitLab、Bitbucket、PyPI 等 截至 2026 年 5 月，有 2880 万个源文件

arXiv:

- Website that allows researchers to share and access papers for free since 1991
- Areas: physics (original), math, CS, statistics, ...
- Has ~3M submissions [article]
- Submission: metadata, PDF, LaTeX source (optional)
- Light approval process (not peer-review)
- Authors choose (i) all rights reserved or (ii) Creative Commons (e.g., CC-BY)
- Metadata (title, abstract) is under a permissive license (CC0)
- Bulk download from Amazon S3, no need to crawl

> arXiv： 自 1991 年起供研究者免费分享和获取论文的网站 领域：物理（最初）、数学、计算机、统计学…… 约有 300 万份投稿 [文章] 提交内容包括：元数据、PDF、LaTeX 源码（可选） 审批流程宽松（不是同行评审） 作者可选择 (i) 保留所有权利，或 (ii) 知识共享许可（如 CC-BY） 元数据（标题、摘要）采用宽松许可（CC0） 可从 Amazon S3 批量下载，无需爬取

[Devlin+ 2018]

The BERT training data consists of:

- Wikipedia
- Books

> BERT 的训练数据由以下构成： 书籍

- Important: sequences are documents rather than sentences
- Contrast: 1 billion word benchmark [Chelba+ 2013] (sentences from machine translation)

> 重要：序列是文档而非句子 对比：十亿词基准 [Chelba+ 2013]（来自机器翻译的句子）

Smashwords

- Founded in 2008, allow anyone to self-publish an e-book
- 2024: 150K authors, 500K books

> 成立于 2008 年，允许任何人自出版电子书 2024 年：15 万名作者、50 万本书

BooksCorpus [Zhu+ 2015]

- Self-published books priced at $0, scraped from Smashwords
- 7K books, 985M words
- Has been taken down because violated Smashwords terms-of-service [article]

> BooksCorpus [Zhu+ 2015] 从 Smashwords 抓取的定价为 $0 的自出版书籍 7000 本书、9.85 亿词 因违反 Smashwords 服务条款而被下架 [文章]

WebText: dataset used to train GPT-2 [Radford+ 2019]

- Contains pages that are outgoing links from Reddit posts with ≥ 3 karma (surrogate for quality)
- 8 million pages, 40GB text

> WebText：用于训练 GPT-2 的数据集 [Radford+ 2019] 包含被 karma ≥ 3 的 Reddit 帖子外链的页面（质量的代理指标） 800 万个页面、40GB 文本

OpenWebTextCorpus: open replication of WebText [Gokaslan+ 2019]

- Extracted all the URLs from the Reddit submissions dataset
- Used Facebook's fastText classifier to filter out non-English
- Removed near duplicates

> OpenWebTextCorpus：WebText 的开放复现 [Gokaslan+ 2019] 从 Reddit submissions 数据集中抽取了全部 URL 用 Facebook 的 fastText 分类器过滤掉非英语内容 移除了近似重复

CCNet [Wenzek+ 2019]

- Goal: automatic way of constructing large, high-quality datasets for pre-training
- Especially interested in getting more data for low-resource languages (e.g., Urdu)

> 目标：一种自动构建大规模、高质量预训练数据集的方法 尤其希望为低资源语言（如乌尔都语）获取更多数据

Components:

- Deduplication: remove duplicate paragraphs based on light normalization
- Language identification: run language ID fastText classifier; keep only target language (e.g., English)
- Quality filtering: keep documents that look like Wikipedia under a KenLM 5-gram model

> 组成部分： 去重：基于轻度规范化删除重复段落 语言识别：运行 fastText 语言识别分类器；只保留目标语言（如英语） 质量过滤：用 KenLM 5-gram 模型保留看起来像 Wikipedia 的文档

Results

- Trained BERT models, CCNet(CommonCrawl) outperforms Wikipedia
- CCNet refers both to the open-source tool and the dataset released from paper

> 结果 训练了 BERT 模型，CCNet（来自 CommonCrawl）优于 Wikipedia CCNet 既指开源工具，也指论文发布的数据集

Colossal Clean Crawled corpus (C4) [Raffel+ 2019]

Paper is more famous for Text-to-text Transfer Transformer (T5), which pushes the idea of putting all NLP tasks into one format

...but a major contribution was the C4 dataset.

> 这篇论文更出名的是 Text-to-Text Transfer Transformer（T5），它推行把所有 NLP 任务统一为一种格式的想法 .....但一项重要贡献是 C4 数据集。

Observation: Common Crawl is mostly not useful natural language

> 观察：Common Crawl 大部分不是有用的自然语言

Started with one snapshot (April 2019) of Common Crawl (1.4 trillion tokens)

> 从 Common Crawl 的一个快照（2019 年 4 月）出发（1.4 万亿 token）

Manual heuristics:

- Keep lines that end in punctuation and have >= 5 words
- Remove page with fewer than 3 sentences
- Removed page that contains any 'bad words' [article]
- Removed page containing '{' (no code), 'lorem ipsum', 'terms of use', etc.

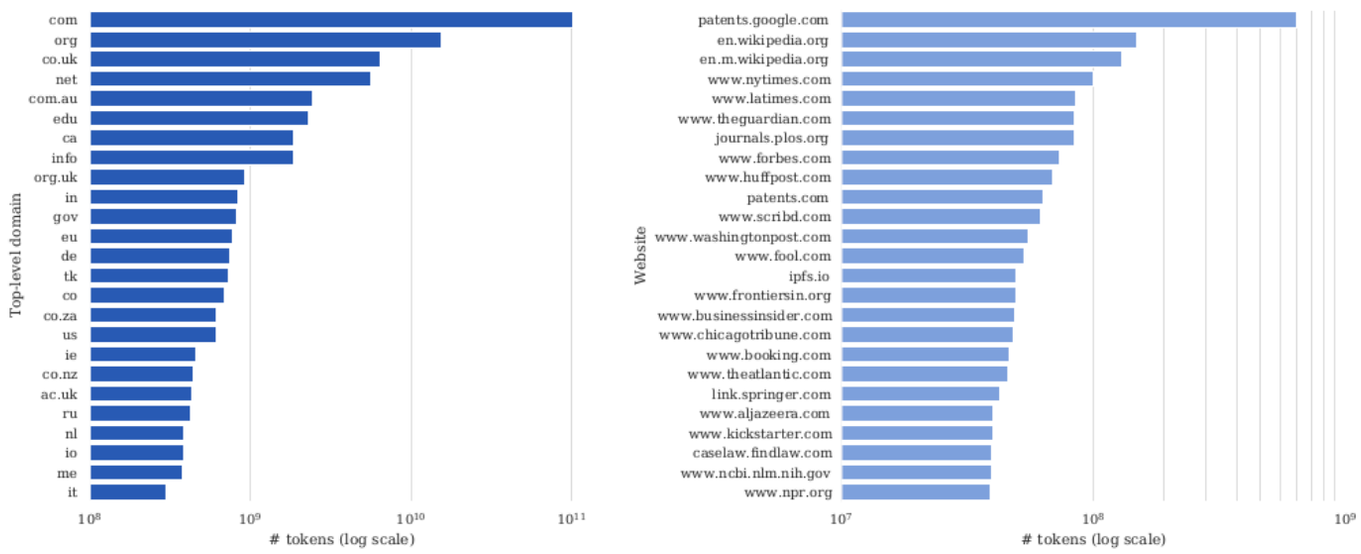
- Filter out non-English text using langdetect (English with probability 0.99)

> 人工启发式规则： 保留以标点结尾且 >= 5 个词的行 删除少于 3 个句子的页面 删除包含任何"脏话"的页面 [文章] 删除包含'{'（排除代码）、'lorem ipsum'、'terms of use' 等的页面 用 langdetect 过滤掉非英语文本（英语概率 0.99）

End result: 806 GB of text (156 billion tokens)

> 最终结果：806 GB 文本（1560 亿 token）

Analysis of C4 [Dodge+ 2021]



> 图：C4 数据集按语言子集的头部域名分布——en、fr、ru、zh 四个面板分别列出该语言子集中占比最高的域名（如 en 面板中的 wikipedia.org 及各类长尾站点）及其百分比。

> 对 C4 的分析 [Dodge+ 2021]

Bonus: WebText-like dataset

- Filtered to pages from OpenWebText links (links in Reddit posts with ≥ 3 karma)
- Used 12 dumps to get 17 GB text (WebText was 40 GB, suggesting CommonCrawl is incomplete)
- This improved on various NLP benchmarks (GLUE, SQuAD, etc.)

> 附加内容：类 WebText 数据集 过滤为来自 OpenWebText 链接（Reddit 帖子中 karma ≥ 3 的链接）的页面 用 12 个转储得到 17 GB 文本（WebText 是 40 GB，说明 CommonCrawl 并不完整） 这在多个 NLP 基准（GLUE、SQuAD 等）上带来了提升

GPT-3 dataset [Brown+ 2020]

- Common Crawl (processed)
- WebText2 (WebText expanded with more links)
- (Mysterious) Internet-based books corpora (Books1, Books2)
- Wikipedia

> GPT-3 数据集 [Brown+ 2020] Common Crawl（经过处理） WebText2（用更多链接扩展的 WebText） （神秘的）基于互联网的书籍语料库（Books1、Books2）

Result: 570 GB (400 billion tokens)

> 结果：570 GB（4000 亿 token）

Common Crawl processing:

- Trained quality classifier to distinguish {WebText, Wikipedia, Books1, Books2} from rest
- Fuzzy deduplication of documents (including WebText and benchmarks)

> Common Crawl 的处理： 训练质量分类器，区分 {WebText, Wikipedia, Books1, Books2} 与其余数据 对文档做模糊去重（包括 WebText 和基准数据）

The Pile [Gao+ 2020]

- In reaction to GPT-3, part of effort to produce open-source language models
- Grassroots effort with lots of volunteers contributing/coordinating on Discord
- Curated 22 high-quality domains

Component	Raw Size	Weight	Epochs	Effective Size	Mean Document Size
Pile-CC	227.12 GiB	18.11%	1.0	227.12 GiB	4.33 KiB
PubMed Central	90.27 GiB	14.40%	2.0	180.55 GiB	30.55 KiB
Books3†	100.96 GiB	12.07%	1.5	151.44 GiB	538.36 KiB
OpenWebText2	62.77 GiB	10.01%	2.0	125.54 GiB	3.85 KiB
ArXiv	56.21 GiB	8.96%	2.0	112.42 GiB	46.61 KiB
Github	95.16 GiB	7.59%	1.0	95.16 GiB	5.25 KiB
FreeLaw	51.15 GiB	6.12%	1.5	76.73 GiB	15.06 KiB
Stack Exchange	32.20 GiB	5.13%	2.0	64.39 GiB	2.16 KiB
USPTO Backgrounds	22.90 GiB	3.65%	2.0	45.81 GiB	4.08 KiB
PubMed Abstracts	19.26 GiB	3.07%	2.0	38.53 GiB	1.30 KiB
Gutenberg (PG-19)†	10.88 GiB	2.17%	2.5	27.19 GiB	398.73 KiB
OpenSubtitles†	12.98 GiB	1.55%	1.5	19.47 GiB	30.48 KiB
Wikipedia (en)†	6.38 GiB	1.53%	3.0	19.13 GiB	1.11 KiB
DM Mathematics†	7.75 GiB	1.24%	2.0	15.49 GiB	8.00 KiB
Ubuntu IRC	5.52 GiB	0.88%	2.0	11.03 GiB	545.48 KiB
BookCorpus2	6.30 GiB	0.75%	1.5	9.45 GiB	369.87 KiB
EuroParl†	4.59 GiB	0.73%	2.0	9.17 GiB	68.87 KiB
HackerNews	3.90 GiB	0.62%	2.0	7.80 GiB	4.92 KiB
YoutubeSubtitles	3.73 GiB	0.60%	2.0	7.47 GiB	22.55 KiB
PhilPapers	2.38 GiB	0.38%	2.0	4.76 GiB	73.37 KiB
NIH ExPorter	1.89 GiB	0.30%	2.0	3.79 GiB	2.11 KiB
Enron Emails†	0.88 GiB	0.14%	2.0	1.76 GiB	1.78 KiB
The Pile	**825.18 GiB**			**1254.20 GiB**	**5.91 KiB**

> 图：The Pile 数据集构成表：列为 Component、Raw Size、Weight、Epochs、Effective Size、Mean Document Size，如 Pile-CC 为 227.12 GiB、权重 18.11%、1 个 epoch。

> 作为对 GPT-3 的回应，是打造开源语言模型努力的一部分 草根项目，大量志愿者在 Discord 上贡献/协作 精心整理了 22 个高质量领域

- 825 GB of text (~275B tokens)
- Pile-CC: Common Crawl, use WARC, jusText to convert into text (better than WET)
- PubMed Central: 5 million papers, mandated to be public for NIH funded work
- arXiv: preprint for research papers since 1991 (use latex)
- Enron emails: 500K emails from 150 users from Enron senior management, released during Enron investigation (2002)

[article]

> 825 GB 文本（约 275B token） Pile-CC：Common Crawl，用 WARC 和 jusText 转成文本（优于 WET） PubMed Central：500 万篇论文，NIH 资助的工作被要求公开 arXiv：自 1991 年起的研究论文预印本（用 latex） 安然邮件：来自安然高管 150 位用户的 50 万封邮件，在安然调查（2002）期间公开 [文章]

Project Gutenberg

- Started in 1971 by Michael Hart, who wanted to increase access to literature
- 2025: ~75K books, mostly English
- Only include books that have received copyright clearance (most in the public domain)

> 由 Michael Hart 于 1971 年创立，希望让文学更容易获得 2025 年：约 7.5 万本书，大部分为英语 只收录已获得版权许可的书籍（大部分处于公有领域）

PG-19: books from Project Gutenberg before 2019 [article]

> PG-19：2019 年之前 Project Gutenberg 的书籍 [文章]

Books3 [Presser, 2020] [article]

- 196K books from the shadow library Bibliotik
- Contained books from authors (e.g., Stephen King, Min Jin Lee, Zadie Smith) [article]
- Has been taken down due to copyright infringement / lawsuits [article]

> 来自影子图书馆 Bibliotik 的 19.6 万本书 包含多位作者的书（如 Stephen King、Min Jin Lee、Zadie Smith） [文章] 因版权侵权 / 诉讼已被下架 [文章]

- Collection of sites of user-contributed questions and answers
- Started with StackOverflow in 2008, grew to other topics (e.g., math, literature) [sites]
- Use reputation points and badges to incentivize participation

- Example

> 一系列用户贡献问答的网站 始于 2008 年的 StackOverflow，后扩展到其他主题（如数学、文学） [站点] 用声望点和徽章激励参与 示例

- Q&A format is close to instruction tuning / real application
- Note: there is metadata (users, votes, comments, badges, tags) for filtering
- Data dumps in XML (anonymized, include metadata) [link]

> 问答格式接近指令微调 / 真实应用 注意：有可用于过滤的元数据（用户、投票、评论、徽章、标签） XML 格式的数据转储（已匿名化，含元数据） [链接]

MassiveText dataset used to train Gopher [Rae+ 2021]
The Gopher model is subsumed by Chinchilla (also never released), but the description of data is good

> 用于训练 Gopher 的 MassiveText 数据集 [Rae+ 2021] Gopher 模型已被 Chinchilla 取代（后者同样从未发布），但其数据描述很值得参考

Components
- MassiveWeb: more on this later
- C4
- Books: no details
- News: no details
- GitHub: no details
- Wikipedia: no details

> 组成部分 MassiveWeb：后面详述 书籍：无细节 新闻：无细节 GitHub：无细节 Wikipedia：无细节

MassiveWeb filtering steps
- Keep English, deduplication, train-test overlap
- Quality filtering using manual rules (not classifier) - e.g., 80% words contain at least one alphabetic character
- Use Google SafeSearch for toxicity (not word lists)

> MassiveWeb 的过滤步骤 保留英语、去重、处理训练-测试重叠 用人工规则做质量过滤（而非分类器）——例如 80% 的词至少含一个字母字符 用 Google SafeSearch 处理毒性内容（而非词表）

Result: 10.5 TB of text (though Gopher only trained on 300B tokens - 12%)

> 结果：10.5 TB 文本（不过 Gopher 只在 300B token 上训练——占 12%）

Dataset for LLaMA [Touvron+ 2023]
- CommonCrawl processed with CCNet, classify *references* of Wikipedia or not
- C4 (more diverse; recall: rule-based filtering)
- GitHub: kept permissive licenses, filtering based on manual rules
- Wikipedia: June-August 2022, 20 languages, manual filtering
- Project Gutenberg and Books3 (from The Pile)
- arXiv: removed comments, inline expanded macros, bibliography
- Stack Exchange: 28 largest websites, sorted answers by score

Result: 1.2T tokens

> LLaMA 的数据集 [Touvron+ 2023] 用 CCNet 处理 CommonCrawl，按是否被 Wikipedia 引用分类 C4（更多样；回顾：基于规则的过滤） GitHub：保留宽松许可证，基于人工规则过滤 Wikipedia：2022 年 6-8 月，20 种语言，人工过滤 Project Gutenberg 和 Books3（来自 The Pile） arXiv：移除注释、内联展开宏、去掉参考文献 Stack Exchange：28 个最大站点，答案按得分排序 结果：1.2T token

Reproduced by Together's RedPajama v1 [https://huggingface.co/datasets/togethercomputer/RedPajama-Data-1T]
Cerebras's SlimPajama: 627B subset of RedPajama v1 by deduplication (MinHashLSH)

> 由 Together 的 RedPajama v1 复现 [https://huggingface.co/datasets/togethercomputer/RedPajama-Data-1T] Cerebras 的 SlimPajama：对 RedPajama v1 去重（MinHashLSH）得到的 627B 子集

RefinedWeb [Penedo+ 2023]
- Point: web data is all you need

- Examples
- trafilatura for HTML→text, extract content (WARC instead of WET files)
- Filtering: Gopher rules, avoid ML-based filtering to avoid biases
- Fuzzy deduplication using MinHash over 5-grams

Released 600B (out of 5T) tokens

> 要点：网络数据就是你所需要的一切 示例 用 trafilatura 做 HTML→文本并抽取正文（用 WARC 而非 WET 文件） 过滤：Gopher 规则；避免基于 ML 的过滤，以免引入偏差 在 5-gram 上用 MinHash 做模糊去重 发布了 600B token（总共有 5T）

FineWeb [article]

- Started as a replication of RefinedWeb, but improved it
- 95 Common Crawl dumps
- URL filtering, language ID (keep if p(en) > 0.65)
- Filtering: Gopher, C4, more manual rules
- Fuzzy deduplication via MinHash
- Anonymize email and public IP addresses (PII)

Result: 15T tokens

> 起初是对 RefinedWeb 的复现，但随后做了改进 95 个 Common Crawl 转储 URL 过滤、语言识别（p(en) > 0.65 时保留） 过滤：Gopher、C4 规则及更多人工规则 通过 MinHash 做模糊去重 对邮箱和公网 IP 地址做匿名化（PII） 结果：15T token

Dolma [Soldaini+ 2024]

Source	Doc Type	UTF-8 bytes *(GB)*	Documents *(millions)*	Unicode words *(billions)*	Llama tokens *(billions)*
Common Crawl	web pages	9,022	3,370	1,775	2,281
The Stack	code	1,043	210	260	411
C4	web pages	790	364	153	198
Reddit	social media	339	377	72	89
PeS2o	STEM papers	268	38.8	50	70
Project Gutenberg	books	20.4	0.056	4.0	6.0
Wikipedia, Wikibooks	encyclopedic	16.2	6.2	3.7	4.3
Total		**11,519**	**4,367**	**2,318**	**3,059**

> 图：Dolma 数据混合构成表：来源包括 Common Crawl（网页，9022GB、33.7 亿文档、22810 亿 Llama tokens）、The Stack（代码）、C4（清洗网页）、Reddit（社交媒体）、PeS2o（STEM 论文）、Project Gutenberg（书籍）、Wikipedia 与 Wikibooks（百科），表列为 UTF-8 字节数、文档数、Unicode 词数与 Llama token 数，总计约 11519GB、30590 亿 tokens。

- Reddit: from the Pushshift project (2005-2023), include submissions and comments separately
- PeS2o: 40M academic papers from Semantic Scholar
- C4, Project Gutenberg, Wikipedia/Wikibooks

> Reddit：来自 Pushshift 项目（2005-2023），帖子和评论分别收录 PeS2o：来自 Semantic Scholar 的 4000 万篇学术论文

Common Crawl processing

- Language identification (fastText classifier), keep English
- Quality filtering (Gopher, C4 rules), avoid model-based filtering
- Toxicity filtering using rules and Jigsaw classifier
- Deduplication using Bloom filters

> Common Crawl 的处理 语言识别（fastText 分类器），保留英语 质量过滤（Gopher、C4 规则），避免基于模型的过滤 用规则和 Jigsaw 分类器做毒性过滤 用 Bloom filter 去重

Result: 3T tokens

> 结果：3T token

DataComp-LM [Li+ 2024]

- Goal: define a standard dataset for trying out different data processing algorithms
- Processed CommonCrawl to produce DCLM-pool (240T tokens)
- DCLM-baseline: filtered down DCLM-pool using quality classifier

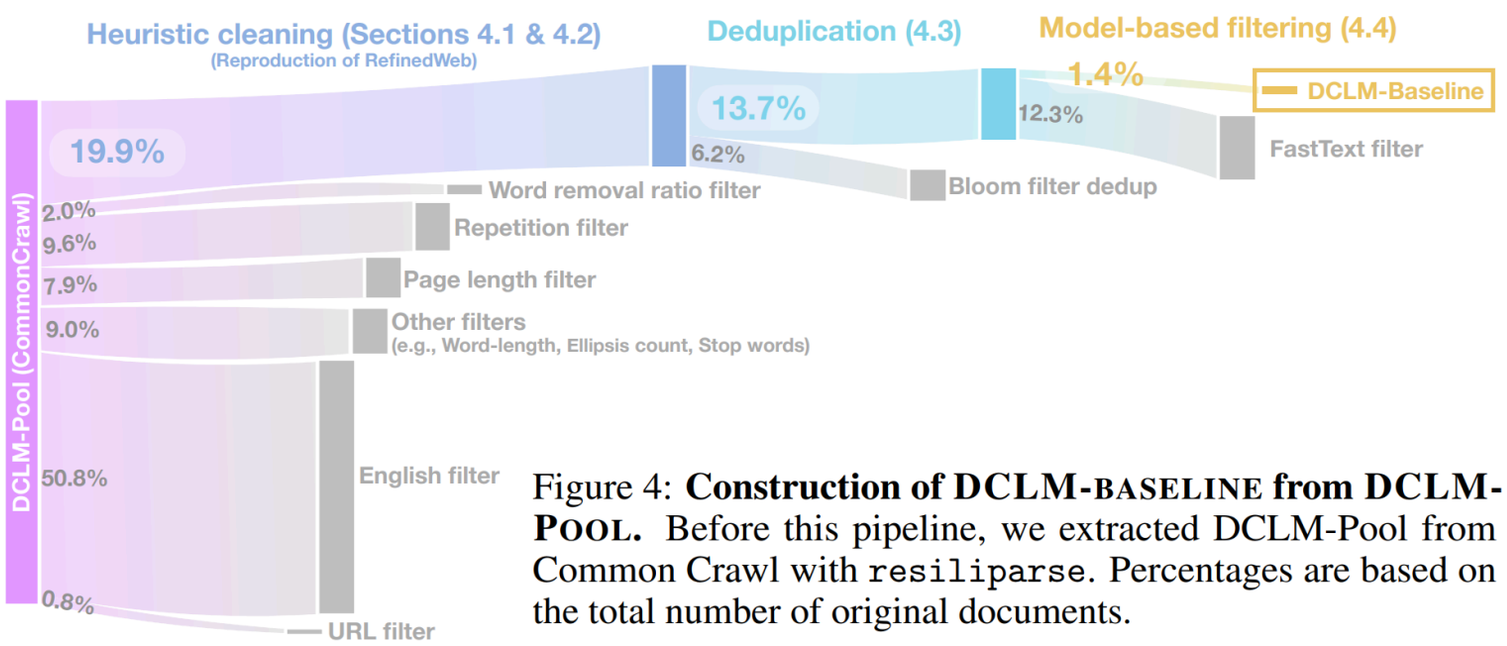


Figure 4: **Construction of DCLM-BASELINE from DCLM-POOL.** Before this pipeline, we extracted DCLM-Pool from Common Crawl with `resiliparse`. Percentages are based on the total number of original documents.

> 图：DCLM 数据清洗流水线：从 DCLM-Pool（CommonCrawl）出发，依次经过启发式清洗（4.1 与 4.2 节，复现 RefinedWeb，含词删除比例过滤、重复过滤等）、去重（4.3 节）与基于模型的过滤（4.4 节）；各阶段标注保留比例（如 19.9%、9.6%、9.0%、50.8%、2.0% 等），展示语料从原始网页逐级收敛到高质量子集。

> 目标：定义一个标准数据集，用于试验不同的数据处理算法 处理 CommonCrawl 得到 DCLM-pool（240T token）
> DCLM-baseline：用质量分类器对 DCLM-pool 过滤缩减

Model-based filtering

Positive examples (200K):

- OpenHermes-2.5: mostly GPT-4 generated instruction data (examples)
- ELI5: subreddit with curiosity questions and answers (examples)

Negative examples (200K):

- RefinedWeb

Result: 3.8T tokens

> 基于模型的过滤 正例（20 万）： OpenHermes-2.5：大部分是 GPT-4 生成的指令数据（示例） ELI5：以好奇心问答为主的 subreddit（示例） 负例（20 万）： 结果：3.8T token

Trained a fastText classifier, run it on all of DCLM-pool

This quality classifier outperforms other filtering methods:

Table 4: **Quality filtering comparison** (`1B-1x` scale). We evaluate various choices for model-based quality filters. Training a `fastText` classifier for filtering performs best.

Filter	CORE	EXTENDED
RefinedWeb reproduction	27.5	14.6
Top 20% by Pagerank	26.1	12.9
SemDedup [1]	27.1	13.8
Classifier on BGE features [185]	27.2	14.0
AskLLM [146]	28.6	14.3
Perplexity filtering	29.0	15.0
Top-k average logits	29.2	14.7
`fastText` [87] OH-2.5 +ELI5	**30.2**	**15.4**

> 图：DCLM 数据池质量过滤示意——原始 Common Crawl 经基于模型的质量过滤（困惑度/质量分类器）后形成精选数据池用于训练，各阶段标注了数据量与留存率。

> 训练了一个 fastText 分类器，在全部 DCLM-pool 上运行 这个质量分类器优于其他过滤方法：

Nemotron-CC [Su+ 2024]

- FineWebEdu and DCLM filter too aggressively (remove 90% of data)

- Need moar tokens (but preserve quality)
- For HTML→text, used jusText (not trafilatura) because it returned more tokens

> FineWebEdu 和 DCLM 过滤得太激进（删掉 90% 的数据） 需要更多 token（但保持质量） HTML→文本转换用了 jusText（而不是 trafilatura），因为它能产出更多 token

Classifier ensembling

- Prompt Nemotron-340B-instruct to score FineWeb documents based on educational value, distill into faster model
- DCLM classifier

> 分类器集成 用 Nemotron-340B-instruct 按教育价值给 FineWeb 文档打分，再蒸馏到更快的模型 DCLM 分类器

Synthetic data rephrasing

- For low-quality data, use LM to rephrase
- For high-quality data, use LM to generate tasks (QA pairs, extract key information, etc.)

> 合成数据改写 对低质量数据，用语言模型改写 对高质量数据，用语言模型生成任务（问答对、抽取关键信息等）

Result: 6.3T tokens (HQ subset is 1.1T)
For reference, Llama 3 trained on 15T, Qwen3 trained on 36T

Dataset	ARC-E	ARC-C	H	W	RACE	PIQA	SIQA	CSQA	OBQA	MMLU	Avg
FineWebEdu-2	71.9	44.7	75.4	67.0	36.8	79.5	45.2	25.5	43.8	42.4	53.2
FineWebEdu	73.6	48.0	70.7	64.6	**38.0**	76.4	43.5	30.0	44.4	42.9	53.2
DCLM	74.7	47.0	76.3	69.1	36.5	79.7	45.6	44.1	44.0	53.4	57.0
Nemotron-CC	75.3	50.7	75.9	67.8	37.9	**80.5**	45.1	47.7	44.2	53.0	57.8
Nemotron-CC-HQ	**78.8**	**52.9**	**76.6**	**69.4**	36.4	80.1	**46.6**	**55.8**	**45.4**	**59.0**	**60.1**

> 图：Nemotron 实验结果表：行为各数据/模型配置，列为 Dataset 及 ARC-E、ARC-C、H、W、RACE、PIQA、SIQA、CSQA、OBQA、MMLU、Avg 等基准得分。

> 结果：6.3T token（高质量子集 1.1T） 作为参考：Llama 3 用 15T 训练，Qwen3 用 36T 训练

The Stack [Kocetkov+ 2022]

- Took repository names from GitHub Archive (2015-2022)
- git clone'd 137M repositories, 51B files (5B unique!)
- Kept only permissively licensed (MIT, Apache) using go-license-detector
- Remove near-duplicates using minhash and Jaccard similarity
- Result: 3.1 TB of code

> 从 GitHub Archive（2015-2022）获取仓库名 git clone 了 1.37 亿个仓库、510 亿个文件（其中仅 50 亿是唯一的！） 用 go-license-detector 只保留宽松许可证（MIT、Apache） 用 minhash 和 Jaccard 相似度去除近似重复 结果：3.1 TB 代码

Stack v2 [Lozhkov+ 2024]

- Issues, comments, PRs from GitHub Archive
- Repositories from the Software Heritage
- Documentation from crawling websites (e.g., PyPI, npm, devdocs.io)
- Processing: remove binary files, malware, bot activity, deduplication, PII redaction, subsample PRs
- Pair source code (especially low-resource languages like Nim) with shared low-level intermediate language (LLVM)
- Include existing datasets (GSM8K, code contests, StackOverflow, arXiv, Wikipedia, OpenWebMath)

> 来自 GitHub Archive 的 issues、评论、PR 仓库来自 Software Heritage 文档来自爬取的网站（如 PyPI、npm、devdocs.io） 处理：移除二进制文件、恶意软件、机器人活动，去重、PII 脱敏、对 PR 采样 把源代码（尤其是 Nim 这类低资源语言）与共享的底层中间语言（LLVM）配对 纳入已有数据集（GSM8K、代码竞赛、StackOverflow、arXiv、Wikipedia、OpenWebMath）

Pull requests:

- Linearize structured object to token sequence
- Add some inline context (e.g., file surrounding diff), subsample

```
<pr>Title: title\nusername_0: description
<pr_status>opened
<repo_name>reponame

<pr_base>
<pr_file>filepath_1
<pr_base_code>file_content/changes_1
...
<pr_file>filepath_N
<pr_base_code>file_content/changes_N

<pr_diff>
<pr_file>filepath_1
<pr_diff_hunk>diff_hunk_1
...
<pr_diff_hunk>diff_hunk_K
...
<pr_file>filepath_M
<pr_diff_hunk>diff_hunk_1
...
<pr_diff_hunk>diff_hunk_J
```

```
<pr_comment>username_id: comment
<pr_event_id>comment_id
...
...
...
<pr_review>username_id: review_comment\n
<pr_event_id>review_id
<pr_review_state>[approved, rejected, commented, changes_required]
...
...
...
<pr_review_comment>
<pr_event_id>comment_id
<pr_in_reply_to_review_id>review_id (opt)
<pr_in_reply_to_comment_id>comment_id (opt)
<pr_file>filepath
<pr_diff_hunk_comment_line>line_number
<pr_diff_hunk>diff_hunk_content
<pr_comment>username_id: comment
```

> 图：Stack v2 中序列化后的 Pull Request 格式示例。左图：PR 头信息与文件改动，用 <pr_status>、<repo_name>、<pr_base>、<pr_file>、<pr_base_code>、<pr_diff>、<pr_diff_hunk> 等标签包裹；右图：PR 评论与评审，用 <pr_comment>、<pr_event_id>、<pr_review>、<pr_review_state>（approved/rejected 等）、<pr_review_comment> 等标签表示，标签名以红色、取值以蓝色标出。

> Pull requests： 将结构化对象线性化为 token 序列 加入一些行内上下文（如 diff 周边的文件内容），并做采样

Recall:

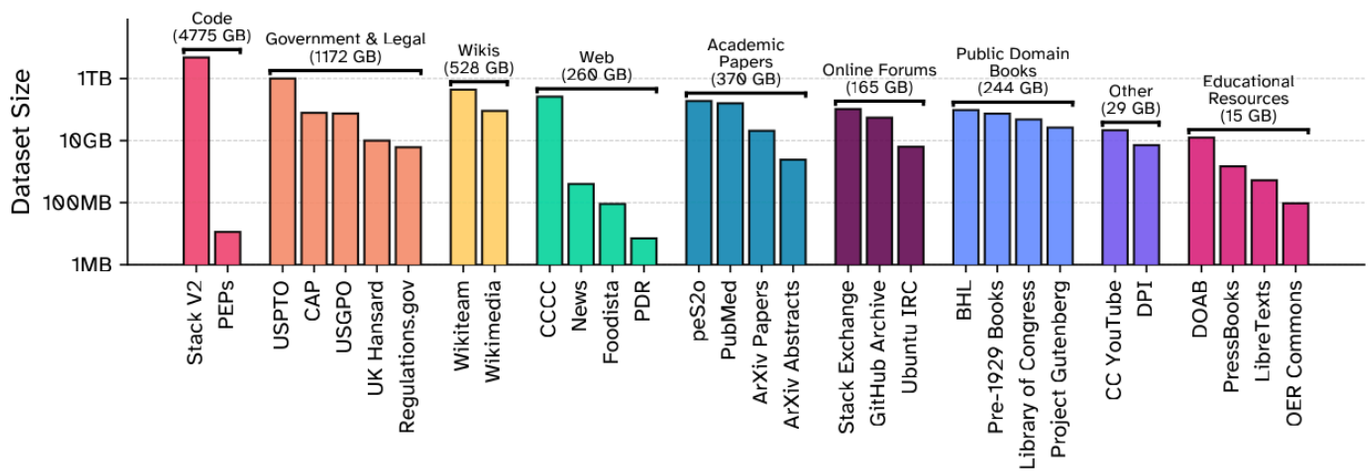
- Almost all data on the Internet is copyrighted.
- Some of it is permissively licensed.
- Fair use of copyrighted content is not settled.

> 回顾： 互联网上几乎所有数据都受版权保护。 其中一部分采用宽松许可。 对版权内容的合理使用尚无定论。

Key question: can you train a good model using only permissively-licensed data?

> 关键问题：只用宽松许可的数据能训练出好模型吗?

CommonPile [Kandpal+ 2025]



> 图：Common Piles 开放语料构成：以数据集大小（1MB–1TB）对各来源作图——代码（Code，4775 GB）、政府与法律（1172 GB）、维基（528 GB）、学术论文（370 GB）、网页（260 GB）、公开领域书籍（244 GB）、网络论坛（165 GB）及其他（29 GB），全部为许可用于训练的开放数据。

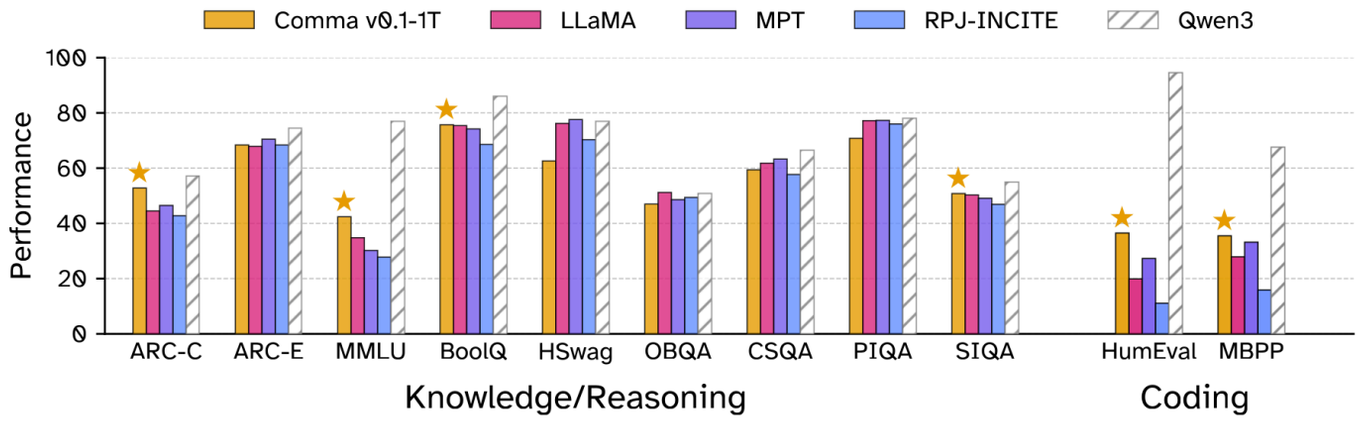
- Collected 8TB dataset of permissively licensed data

> 收集了 8TB 的宽松许可数据集

Subtleties:

- License laundering: redistribute copyrighted work under permissive license (hard to detect)
- Collection licenses (Dolma is ODC-By) doesn't extend to individual
- Synthetic data from LMs trained on unlicensed data is unclear

> 一些微妙之处： 许可洗白：以宽松许可再分发受版权保护的作品（很难检测） 数据集整体的许可（Dolma 是 ODC-By）并不延伸到其中的单个内容 由在未授权数据上训练的语言模型生成的合成数据，法律地位不明



> 图：comma v0.1 的质量与数量实验结果——左图：AI2 Reasoning Challenge 准确率（%）随训练 token 数（B）的变化；右图：MMLU（%）随训练 token 数的变化；曲线对比 comma 基线与 comma v0.1 在 3B–36B token 等检查点上的提升。

- Can do decently, but tough to compete without more tokens

> 能做得还不错，但缺更多 token 就很难有竞争力